

module5_lab

October 4, 2026

```
[1]: import pandas as pd
df=pd.read_csv("Downloads/chickwts.csv")
```

```
[2]: df.head
```

```
[2]: <bound method NDFrame.head of      Unnamed: 0  weight      feed
0           1      179  horsebean
1           2      160  horsebean
2           3      136  horsebean
3           4      227  horsebean
4           5      217  horsebean
..          ...      ...      ...
66          67      359    casein
67          68      216    casein
68          69      222    casein
69          70      283    casein
70          71      332    casein

[71 rows x 3 columns]>
```

```
[3]: df.shape
```

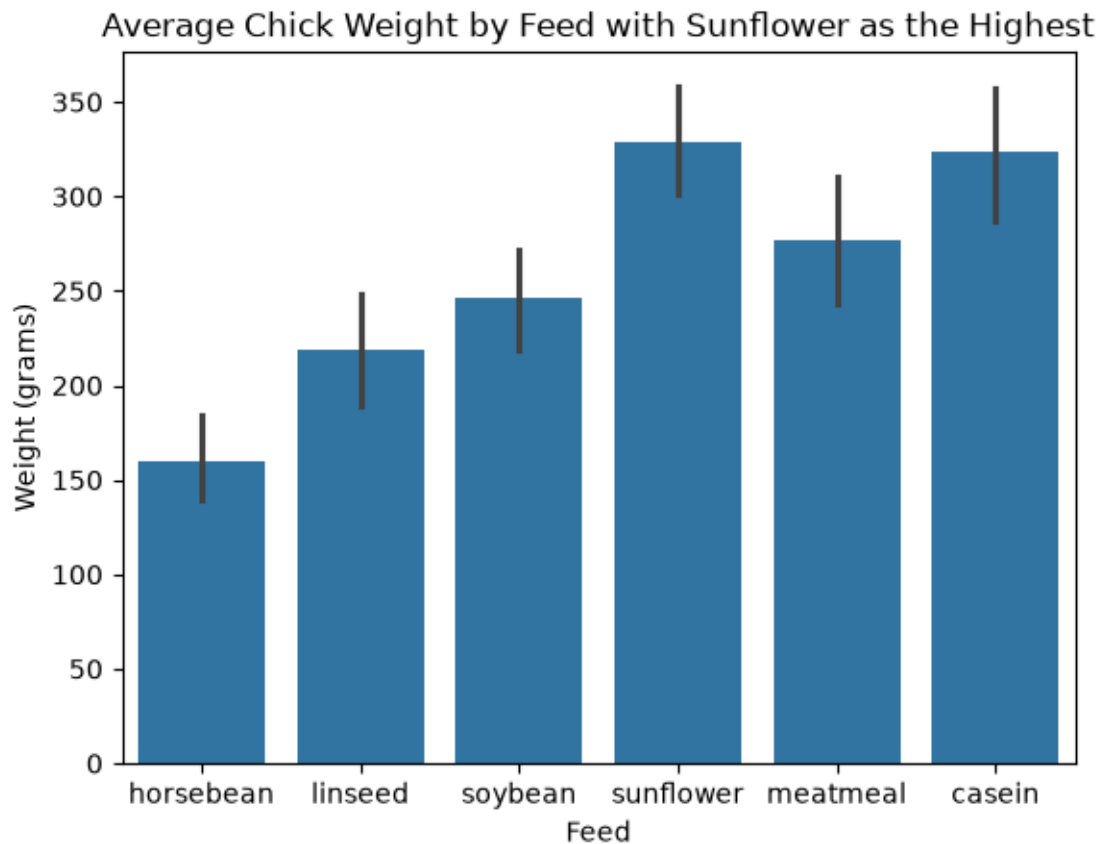
```
[3]: (71, 3)
```

```
[4]: import seaborn as sns
import matplotlib.pyplot as plt
```

```
[5]: df.groupby("feed")["weight"].mean().sort_values()
```

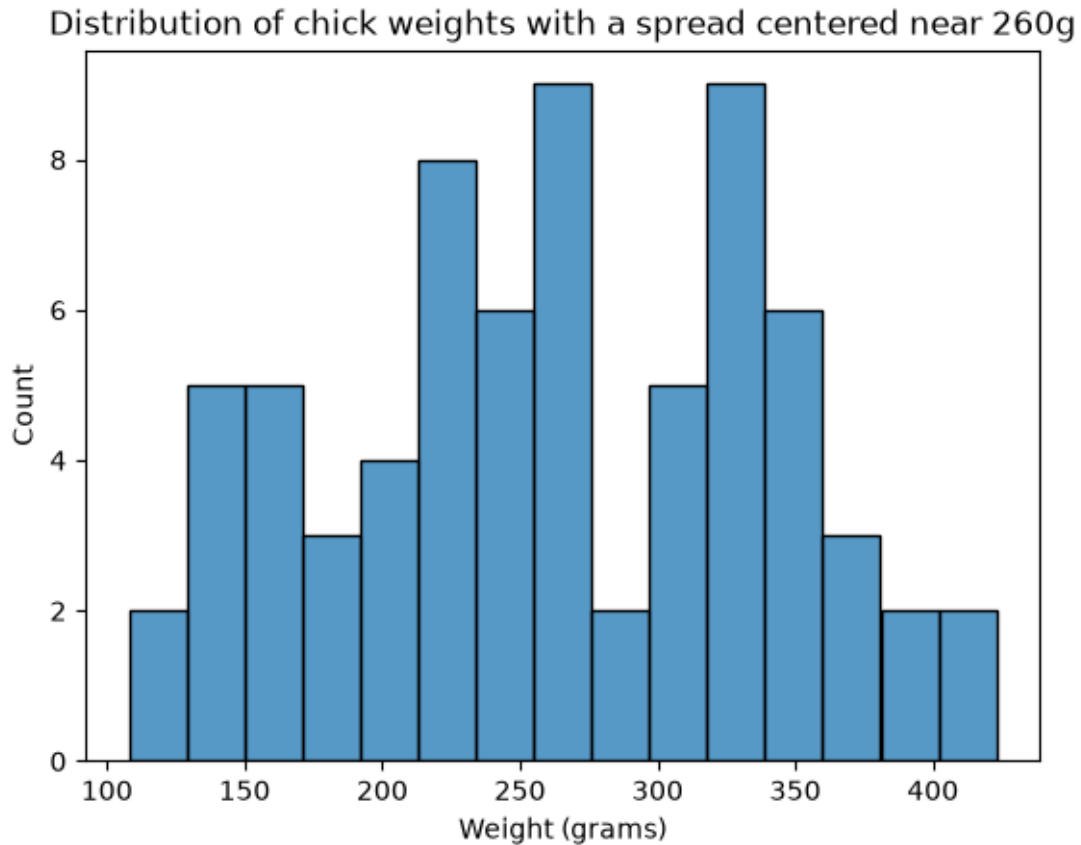
```
[5]: feed
horsebean    160.200000
linseed      218.750000
soybean      246.428571
meatmeal     276.909091
casein       323.583333
sunflower    328.916667
Name: weight, dtype: float64
```

```
[6]: sns.barplot(data=df,x="feed",y="weight")
plt.title("Average Chick Weight by Feed with Sunflower as the Highest")
plt.xlabel("Feed")
plt.ylabel("Weight (grams)")
plt.show()
```



```
[7]: sns.histplot(data=df, x="weight", bins=15)
plt.title("Distribution of chick weights with a spread centered near 260g")
plt.xlabel("Weight (grams)")
plt.ylabel("Count")
```

```
[7]: Text(0, 0.5, 'Count')
```



```
[8]: df=pd.read_csv("Downloads/museum_visitors.csv")
```

```
[9]: df.shape
```

```
[9]: (59, 5)
```

```
[10]: df.head
```

```
[10]: <bound method NDFrame.head of
Chinese American Museum \
0    2014-01-01    24778    4486    1581
1    2014-02-01    18976    4172    1785
2    2014-03-01    25231    7082    3229
3    2014-04-01    26989    6756    2129
4    2014-05-01    36883   10858    3676
5    2014-06-01    29487    5751    2121
6    2014-07-01    32378    5406    2239
7    2014-08-01    37680    8619    1769
8    2014-09-01    28473   61192    1073
9    2014-10-01    27995    6488    1979
```

10	2014-11-01	25691	4189	2404
11	2014-12-01	18754	4339	1319
12	2015-01-01	20438	3858	1823
13	2015-02-01	15578	3742	1558
14	2015-03-01	21297	5390	2336
15	2015-04-01	26670	7000	3057
16	2015-05-01	34383	12528	4009
17	2015-06-01	30569	5377	2544
18	2015-07-01	41242	6111	3057
19	2015-08-01	30700	5383	2415
20	2015-09-01	20967	5746	1398
21	2015-10-01	29764	8882	2237
22	2015-11-01	24483	6848	2850
23	2015-12-01	21426	4502	2075
24	2016-01-01	19659	4377	2150
25	2016-02-01	17378	3675	2547
26	2016-03-01	20322	6334	7702
27	2016-04-01	24521	6509	3485
28	2016-05-01	31728	9315	4565
29	2016-06-01	23696	4918	3034
30	2016-07-01	31689	5602	2776
31	2016-08-01	25987	5181	2153
32	2016-09-01	22897	4736	1754
33	2016-10-01	25040	6145	2607
34	2016-11-01	17760	3930	2248
35	2016-12-01	20107	3911	2263
36	2017-01-01	18792	3551	2531
37	2017-02-01	14035	3491	2139
38	2017-03-01	20680	5273	5966
39	2017-04-01	25234	5320	3273
40	2017-05-01	30029	8733	5585
41	2017-06-01	22169	4862	2946
42	2017-07-01	30831	5242	2939
43	2017-08-01	27009	5379	1784
44	2017-09-01	23403	4484	1704
45	2017-10-01	22164	4822	2419
46	2017-11-01	17629	4613	2685
47	2017-12-01	18339	4448	2768
48	2018-01-01	16265	3910	2477
49	2018-02-01	14718	3306	2710
50	2018-03-01	19001	5225	4529
51	2018-04-01	18966	4535	2750
52	2018-05-01	25173	8013	4960
53	2018-06-01	22171	4635	2518
54	2018-07-01	23136	4191	2620
55	2018-08-01	20815	4866	2409
56	2018-09-01	21020	4956	2146

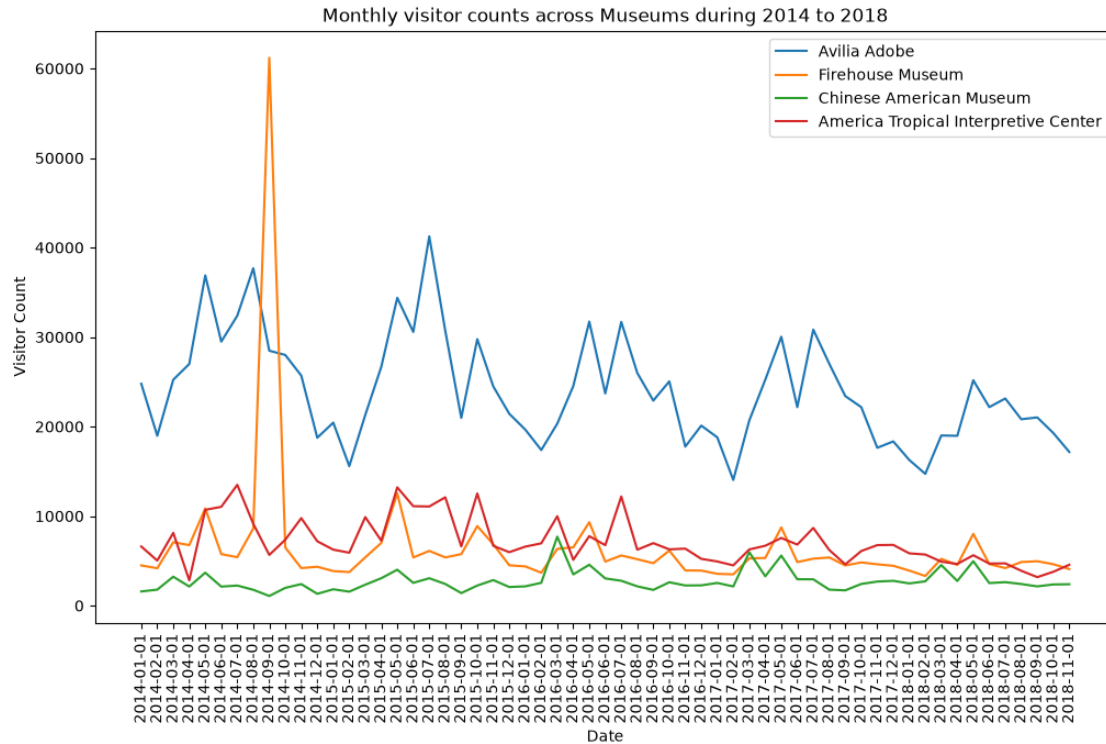
57	2018-10-01	19280	4622	2364
58	2018-11-01	17163	4082	2385

America Tropical Interpretive Center

0	6602
1	5029
2	8129
3	2824
4	10694
5	11036
6	13490
7	9139
8	5661
9	7356
10	9773
11	7184
12	6250
13	5907
14	9884
15	7254
16	13207
17	11102
18	11072
19	12096
20	6608
21	12524
22	6677
23	5967
24	6587
25	6955
26	9979
27	5113
28	7757
29	6751
30	12188
31	6245
32	6977
33	6282
34	6365
35	5223
36	4929
37	4493
38	6271
39	6685
40	7556
41	6818
42	8675

43	6212
44	4607
45	6097
46	6757
47	6780
48	5837
49	5705
50	4909
51	4650
52	5626
53	4694
54	4718
55	3891
56	3180
57	3775
58	4562 >

```
[11]: plt.figure(figsize=(12,7))
sns.lineplot(data=df, x="Date", y="Avila Adobe", label="Avilia Adobe")
sns.lineplot(data=df, x="Date", y="Firehouse Museum",label="Firehouse Museum")
sns.lineplot(data=df, x="Date", y="Chinese American Museum",label="Chinese_
↳American Museum")
sns.lineplot(data=df, x="Date", y="America Tropical Interpretive_
↳Center",label="America Tropical Interpretive Center")
plt.title("Monthly visitor counts across Museums during 2014 to 2018")
plt.xlabel("Date")
plt.ylabel("Visitor Count")
plt.legend()
plt.xticks(rotation=90)
plt.show()
```

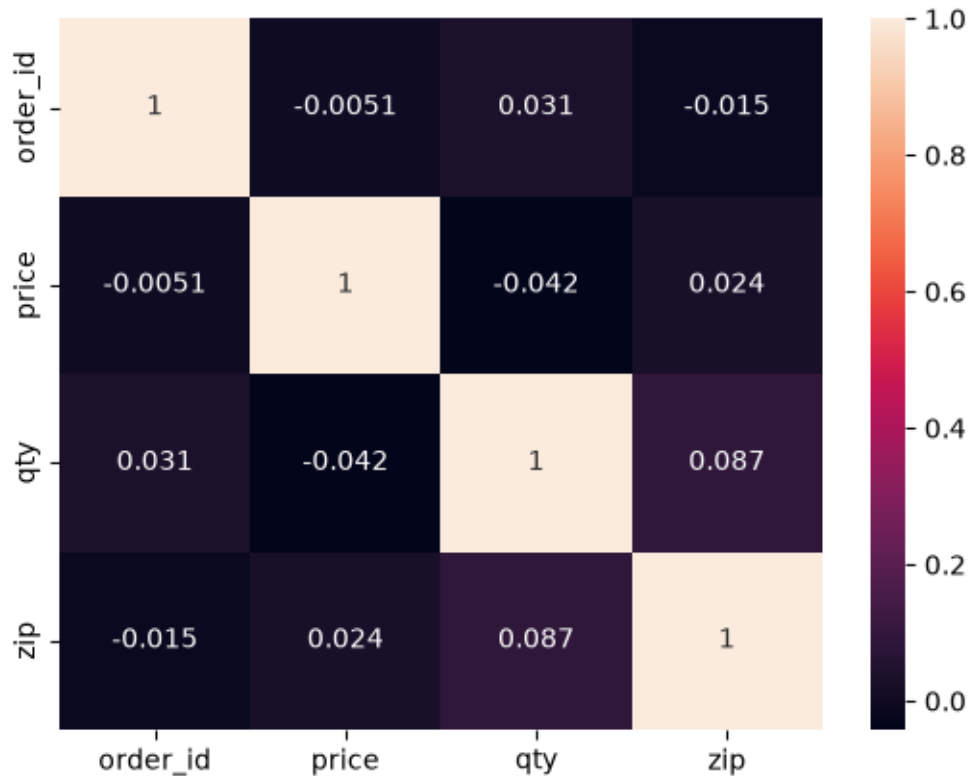


```
[12]: #markdown cell
#there is a rises and dips but the one for Firehouse Museum had a huge spike in
↳late 2014
#Honest-chart audit
# For the monthly museum visitor chart 2 design choices for data honesty was
↳the use of a 0 baseline for the scale of visitors. Second, was plotting the
↳timeline from 4014 to 2018 with no ommited months.
```

```
[13]: df=pd.read_csv("Downloads/sales_clean.csv")
```

```
[14]: sns.heatmap(df.corr(numeric_only=True),annot=True)
```

```
[14]: <Axes: >
```



```
[15]: #the largest off correlation is between quantity and zip code at .087
```

```
[16]: #Null Hypothesis - There is no significant difference in weight between
      ↪sunflower fed chicks and horsebean fed chicks.
```

```
[17]: df=pd.read_csv("Downloads/chickwts.csv")
      from scipy.stats import ttest_ind
      a = df[df["feed"] == "sunflower"]["weight"]
      b = df[df["feed"] == "horsebean"]["weight"]
      ttest_ind(a, b)
```

```
[17]: TtestResult(statistic=np.float64(8.848346742516636),
      pvalue=np.float64(2.3743507676837164e-08), df=np.float64(20.0))
```

```
[18]: #Chicks fed sunflower meal weighed an average of 168.7 grams more than those
      ↪fed horsebean the p value being 2.3743507676837164e-08, leading to the
      ↪rejection of the null hypothesis, this finding should be used with caution
      ↪given the sample size.
```